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PHYS 537

Problem 1 – Variational Ground State in a One-Dimensional Box

1. The nonorthogonal Rayleigh-Ritz basis functions are

$$b_j = x(1-x)x^j, j = 0, 1, \dots, M$$

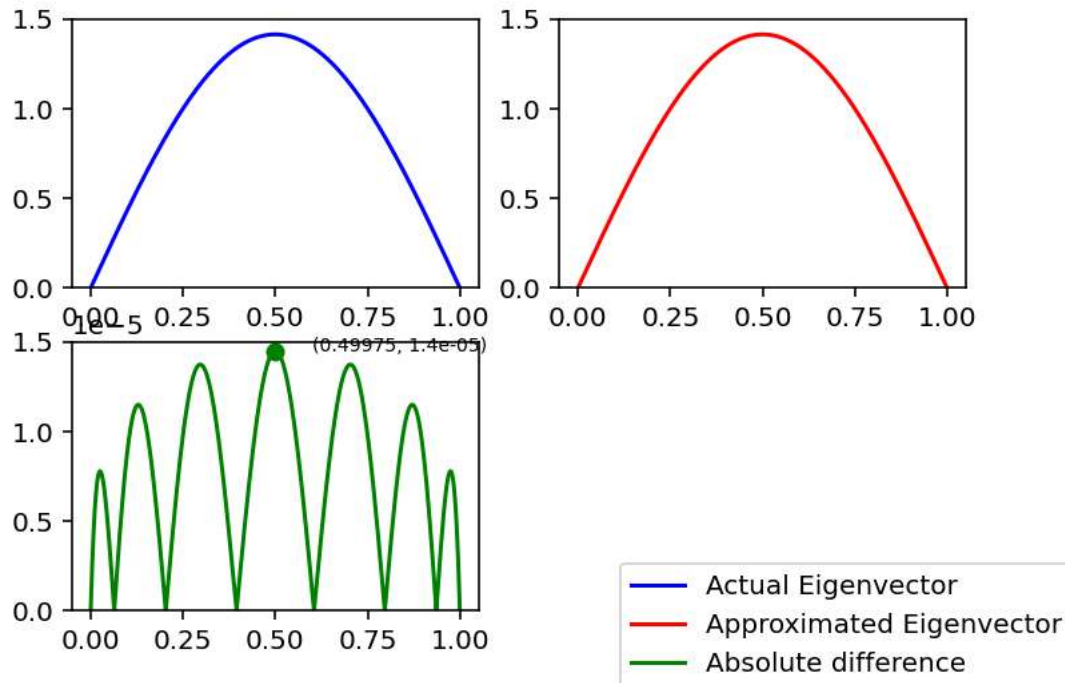
The boundary conditions for this problem are

$$\psi(0) = \psi(1) = 0$$

It can very easily be shown that each basis function satisfies both boundary conditions.

Substituting zero in for either x or x^j will make $b_j = 0$, and inserting one into $1 - x$ will do the same. A generalized eigenvalue problem is needed because of the nonorthogonality of basis functions and the fact that many of them might have the same eigenvalue.

2. Done
3. Values in the console. We can see that that the lowest eigenvalue from the approximation is about twice as large as the exact dimensionless ground-state energy. All calculated eigenvalues from this method are positive and larger than the actual eigenvalue, so this satisfies the variational upper-bound property that all eigenvalues must serve as an upper bound on the actual ground state energy eigenvalue. The method used for calculating the eigen-vectors/-values automatically sorts the eigenvalues from least to greatest and also the eigenvectors in to position based on respective eigenvalue order. For the reason, and also that eigenvalues and eigenvectors can be calculated in any order, it's hard to say it they're monotonically nonincreasing. If I use the sequence that's immediately available from the calculation, they would be monotonically increasing.
4. These are my graphs of the estimated wavefunction, the actual wavefunction and the absolute difference between them, with the maximum labeled:



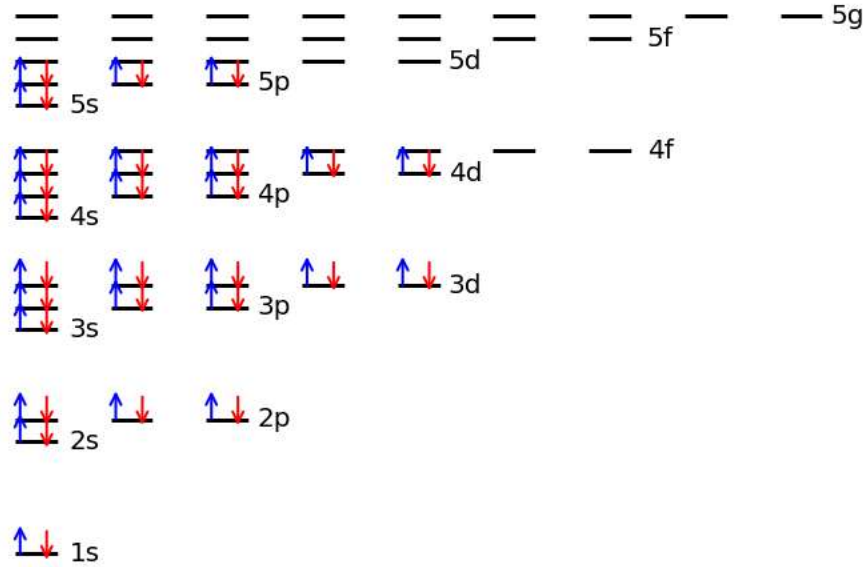
The two eigenvector plots are separated because when I put them in the same plot, they overlap seemingly perfectly (despite what the green graph would suggest). The absolute difference graph includes the point of maximum absolute difference. I couldn't move the point label without breaking Spyder's display feature for some reason, so it's a bit hard to see. The y-scale of that graph is on the order of 10^{-5} and the point of interest is (0.49975, 1.4×10^{-5}).

5. In a generalized eigenvalue problem, there is room for many linearly independent eigenfunctions to correspond to the same eigenvalue. Adding a basis vector may allow for these other eigenfunctions to be measured. However, since they add no new energies, the lowest energy calculated would be left unchanged.

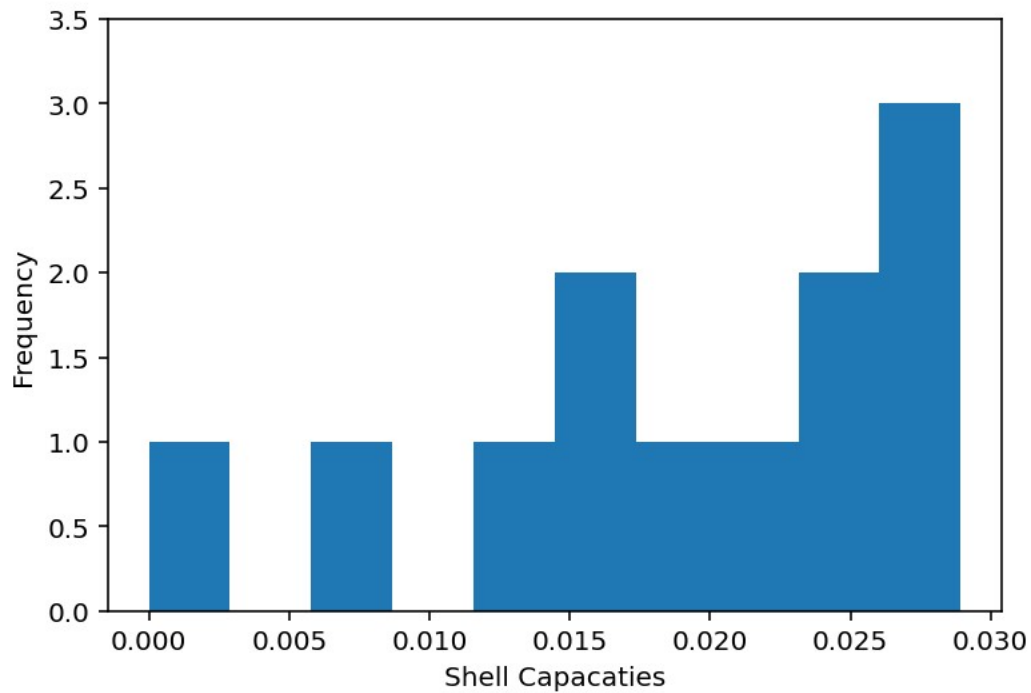
Problem 2 – Periodic State Counting and Fermi Filling

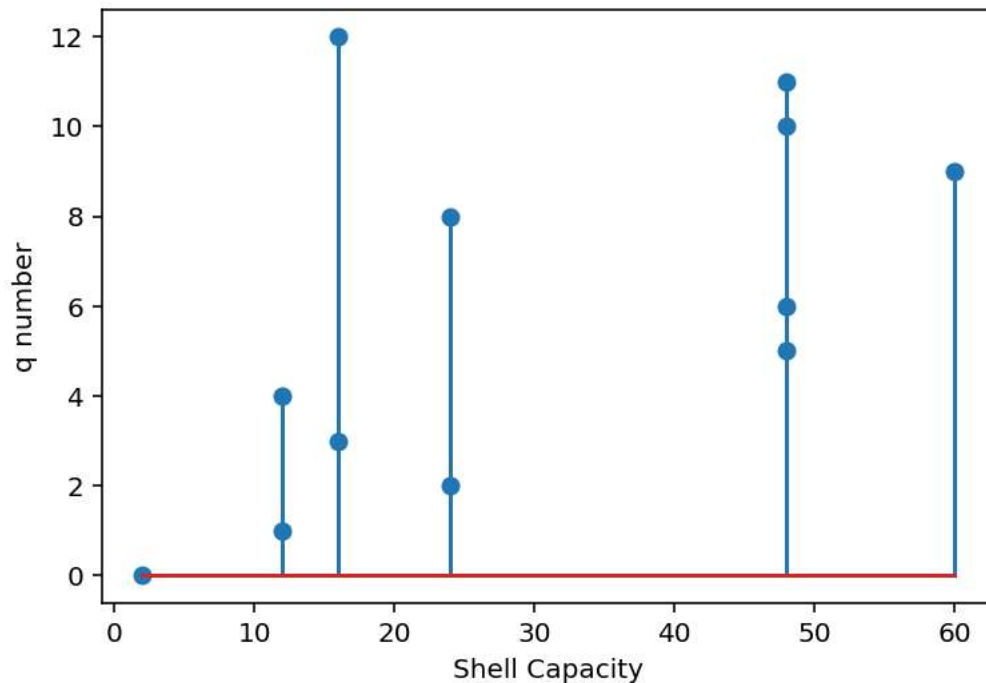
1. Enumeration in the console.

Filling Orbitals for 54 Electrons



- 2.
3. These value are given in the console. The percent difference of the continuum Fermi energy from the finite-cell energy is about 95.4558%, which comes from not considering the factor of q .

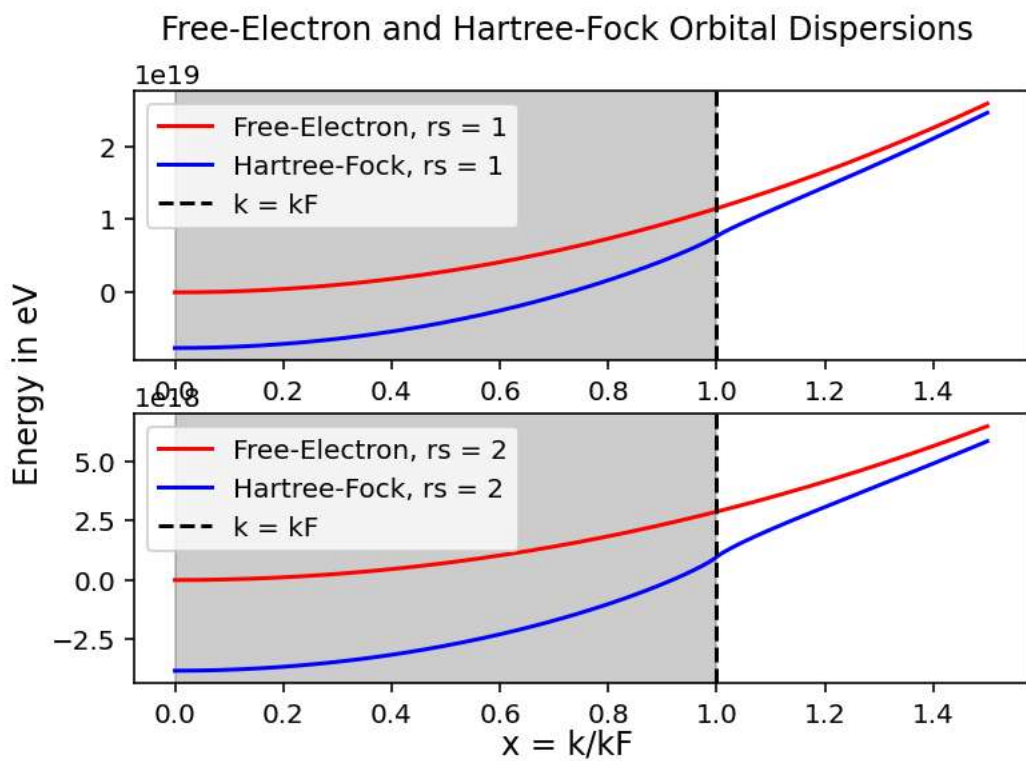
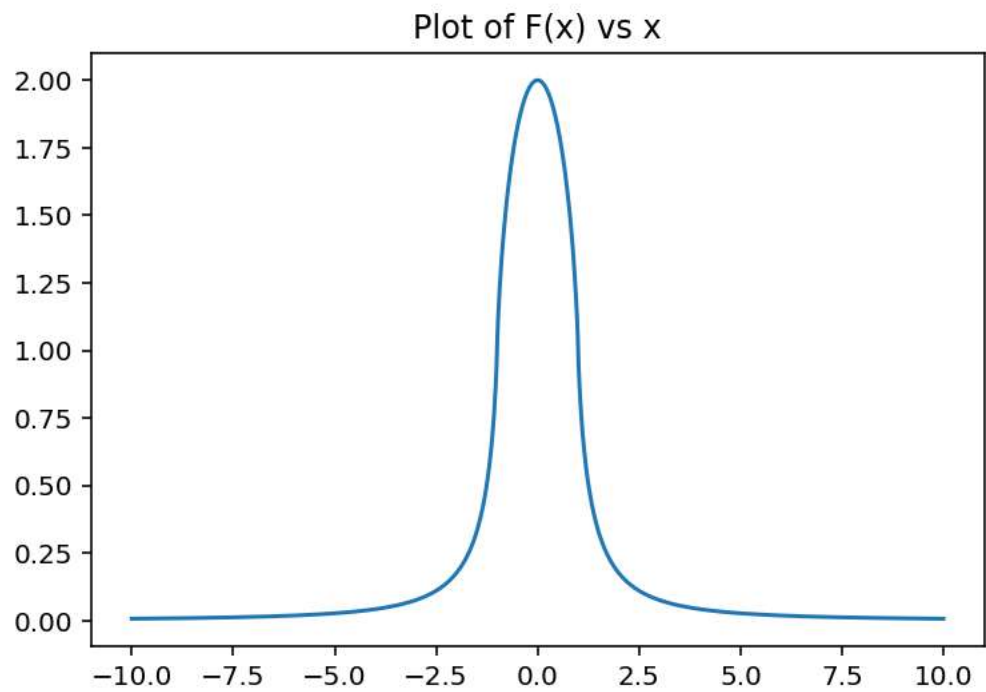




- 4.
5. When it comes to the electrons in an orbital, there are always twice as many electrons as orbitals thanks to the spin-inclusive capacity. Even for $q = 0$, there is one spatial orbital that accommodates two electrons on its own. So, from the start, spin is taken into account. The wavenumbers, however, are allowed to take a positive or negative value due to the quadratic relationship Energy has with it. That said, when $q = 0$, k can only have one value – zero – so the positive and negative k contributions enter the counting after $q = 0$. These factors don't enter twice because that would require quartic relationships where we only see quadratic relationships.

Problem 3 – Hartree-Fock Exchange in Jellium

1. The limiting cases in my code are handled by a piecewise function that fills in the values that weren't allowed by the rest of the function. The rest of the function when plotted on large enough scale won't get a divide by zero or log of zero error because the plotter just ignores them. Analytically, though, these limiting values can be calculated by – and hold onto your hat because this might be a shocker – taking a limit of F as x goes to a bad value. The value of those limits are what replace the divide by zero and log of zero errors and make the function continuous. It turns out that those x -values are points of removable discontinuity.



2.

	kF	eFree	eHF	avg Kinetic	exchange E per e-	EHF/N
$r_s = 1$	1.919158e+00	1.149517e+19	7.682012e+18	6.897101e+18	-2.859868e+18	4.037234e+18
$r_s = 2$	9.595791e-01	2.873792e+18	9.672138e+17	1.724275e+18	-1.429934e+18	2.943415e+17

- 3.
4. The symmetric differences are reported in the console as an array of arrays, each containing the calculated symmetric differences as they increment closer to $x = 1$. The arrays inside the more global array are ordered from largest delta to smallest delta. They show that with smaller delta, the symmetric difference close to $x = 1$ increases. The trend seems to increase this number by about 0.7 for each next smaller delta. So the differences seem to be growing with decreasing order of magnitude of delta. This is a divergence on a logarithmic scale, and suggests an asymptotically increasing slope at the Fermi wavenumber. Since density of states is inversely related to this slope, and infinitely increasing slope would cause the density of states to go to zero at the Fermi wavenumber, which just isn't physical.
5. exchange term is nonlocal because it isn't a function of position; it's a function of wavenumber, which is calculated from the principal quantum numbers, which themselves relate to energy levels. Planewaves are still eigenfunctions in homogeneous systems for the same reason – the exchange term is nonlocal due to the uniform background that replaces the positive ions, so it has translational symmetry, allowing the plane waves to remain eigenfunctions as the operator commutes with translations. The exchange operator also only has contributions between same-spin orbitals due to the orthogonality of opposite sign functions under the defining integral.